

Questions

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Q1 - 2024 (04 Apr Shift 1)

If 2 and 6 are the roots of the equation $ax^2 + bx + 1 = 0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is :

(1) $2x^2 + 11x + 12 = 0$

(2) $x^2 + 8x + 12 = 0$

(3) $4x^2 + 14x + 12 = 0$

(4) $x^2 + 10x + 16 = 0$

Q2 - 2024 (05 Apr Shift 2)

The number of real solutions of the equation $x|x+5| + 2|x+7| - 2 = 0$ is _____

Q3 - 2024 (06 Apr Shift 1)

Let α, β be the distinct roots of the equation $x^2 - (t^2 - 5t + 6)x + 1 = 0, t \in \mathbb{R}$ and $a_n = \alpha^n + \beta^n$. Then the minimum value of $\frac{a_{2023} + a_{2025}}{a_{2024}}$ is

(1) $-1/4$

(2) $-1/4$

(3) $-1/2$

(4) $1/4$

Q4 - 2024 (06 Apr Shift 1)

Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and

$(4 + x_1^2)(4 + x_2^2)(4 + x_3^2)(4 + x_4^2) = \frac{125}{16}m$. Then the value of m is

Q5 - 2024 (06 Apr Shift 2)

Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ is equal to _____

Q6 - 2024 (08 Apr Shift 1)

Questions

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The sum of all the solutions of the equation $(8)^{2x} - 16 \cdot (8)^x + 48 = 0$ is :

(1) $1 + \log_8(6)$

(2) $1 + \log_6(8)$

(3) $\log_8(6)$

(4) $\log_8(4)$

Q7 - 2024 (08 Apr Shift 2)

The number of distinct real roots of the equation $|x + 1| + |x + 3| - 4|x + 2| + 5 = 0$, is

Q8 - 2024 (09 Apr Shift 1)

Let α, β be the roots of the equation $x^2 + 2\sqrt{2}x - 1 = 0$. The quadratic equation, whose roots are $\alpha^4 + \beta^4$ and $\frac{1}{10}(\alpha^6 + \beta^6)$, is :

(1) $x^2 - 190x + 9466 = 0$

(2) $x^2 - 180x + 9506 = 0$

(3) $x^2 - 195x + 9506 = 0$

(4) $x^2 - 195x + 9466 = 0$

Q9 - 2024 (09 Apr Shift 2)

Let $\alpha, \beta; \alpha > \beta$, be the roots of the equation $x^2 - \sqrt{2}x - \sqrt{3} = 0$. Let $P_n = \alpha^n - \beta^n, n \in \mathbb{N}$. Then $(11\sqrt{3} - 10\sqrt{2})P_{10} + (11\sqrt{2} + 10)P_{11} - 11P_{12}$ is equal to

(1) $10\sqrt{3}P_9$

(2) $11\sqrt{3}P_9$

(3) $10\sqrt{2}P_9$

(4) $11\sqrt{2}P_9$

Questions

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Answer Key

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Solutions

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Q1

$$\text{Sum} = 8 = -\frac{b}{a}$$

$$\text{Product} = 12 = \frac{1}{a} \Rightarrow a = \frac{1}{12}$$

$$b = -\frac{2}{3}$$

$$2a + b = \frac{2}{12} - \frac{2}{3} = -\frac{1}{2}$$

$$6a + b = \frac{6}{12} - \frac{2}{3} = -\frac{1}{6}$$

$$\text{sum} = -8$$

$$P = 12$$

$$x^2 + 8x + 12 = 0$$

Q2

Case I : $x \geq -5$

$$x^2 + 5x + 2x + 12 = 0$$

$$x^2 + 7x + 12 = 0$$

$$x = -3, -4$$

Case II : $-7 < x < -5$

$$-x^2 - 5x + 2x + 14 - 2 = 0$$

$$-x^2 - 3x + 12 = 0$$

$$-3 \pm \sqrt{9 + 48}$$

$$x = \frac{-3 \pm \sqrt{57}}{2}$$

$$= \frac{-3 \pm \sqrt{57}}{2}$$

$$x = \frac{-3 - \sqrt{57}}{2}, \frac{-3 + \sqrt{57}}{2} \text{ (rejected)}$$

Case III : $x \leq -7$

$$-x^2 - 5x - 2x - 14 - 2 = 0$$

$$x^2 + 7x + 16 = 0$$

$$D = 49 - 64 < 0$$

No solutions

No. of solutions = 3

Q3

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Solutions

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by newton's theorem

$$a_{n+2} - (t^2 - 5t + 6) a_{n+1} + a_n = 0$$

$$\therefore a_{2025} + a_{2023} = (t^2 - 5t + 6) a_{2024}$$

$$\therefore \frac{a_{2025} + a_{2023}}{a_{2024}} = t^2 - 5t + 6$$

$$\therefore t^2 - 5t + 6 = \left(t - \frac{5}{2}\right)^2 - \frac{1}{4}$$

$$\therefore \text{minimum value} = -\frac{1}{4}$$

Q4

$$4x^4 + 8x^3 - 17x^2 - 12x + 9$$

$$= 4(x - x_1)(x - x_2)(x - x_3)(x - x_4)$$

Put $x = 2i$ & $-2i$

$$64 - 64i + 68 - 24i + 9 = (2i - x_1)(2i - x_2)(2i - x_3)(2i - x_4)$$

$$= 141 - 88i \dots (1)$$

$$64 + 64i + 68 + 24i + 9 = 4(-2i - x_1)(-2i - x_2)(-2i - x_3)(-2i - x_4)$$

$$= 141 + 88i \dots (2)$$

$$\frac{125}{16} m = \frac{141^2 + 88^2}{16}$$

$$m = 221$$

Q5

$$\frac{\alpha^{10} + \beta^{10} + \sqrt{2}(\alpha^9 + \beta^9)}{2(\alpha^8 + \beta^8)}$$

$$\frac{\alpha^8(\alpha^2 + \sqrt{2}\alpha) + \beta^8(\beta^2 + \sqrt{2}\beta)}{2(\alpha^8 + \beta^8)}$$

$$\frac{8\alpha^8 + 8\beta^8}{2(\alpha^8 + \beta^8)} = 4$$

Q6

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Solutions

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$$(8)^{2x} - 16 \cdot (8)^x + 48 = 0$$

$$\text{Put } 8^x = t$$

$$t^2 - 16 + 48 = 0$$

$$\Rightarrow t = 4 \text{ or } t = 12$$

$$\Rightarrow 8^x = 4 \quad 8^x = 12$$

$$\Rightarrow x = \log_8 4 \quad x = \log_8 12$$

$$\text{sum of solution} = \log_8 4 + \log_8 12$$

$$= \log_8 48 = \log_8 (6 \cdot 8)$$

$$= 1 + \log_8 6$$

Q7

$$|x+1||x+3| - 4|x+2| + 5 = 0$$

case-1

$$x \leq -3$$

$$(x+1)(x+3) + 4(x+2) + 5 = 0$$

$$x^2 + 4x + 3 + 4x + 8 + 5 = 0$$

$$x^2 + 8x + 16 = 0$$

$$(x+4)^2 = 0$$

$$x = -4$$

case-2

$$-3 \leq x \leq -2$$

$$-x^2 - 4x - 3 + 4x + 8 + 5 = 0$$

$$-x^2 + 10 = 0$$

$$x = \pm\sqrt{10}$$

case-3

$$-2 \leq x \leq -1$$

$$-x^2 - 4x - 3 - 4x - 8 + 5 = 0$$

$$-x^2 - 8x - 6 = 0$$

$$x^2 + 8x + 6 = 0$$

$$x = \frac{-8 \pm 2\sqrt{10}}{2} = -4 \pm \sqrt{10}$$

case-4

$$x \geq -1$$

$$x^2 + 4x + 3 - 4x - 8 + 5 = 0$$

$$x^2 = 0$$

$$x = 0$$

$$\text{No. of solution} = 2$$

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Solutions

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Q8

$$x^2 + 2\sqrt{2}x - 1 = 0$$

$$\alpha + \beta = -2\sqrt{2}$$

$$\alpha\beta = -1$$

$$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$$

$$= ((\alpha + \beta)^2 - 2\alpha\beta)^2 - 2(\alpha\beta)^2$$

$$= (8 + 2)^2 - 2(-1)^2$$

$$= 100 - 2 = 98$$

$$\alpha^6 + \beta^6 = (\alpha^3 + \beta^3)^2 - 2\alpha^3\beta^3$$

$$= ((\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta))^2 - 2(\alpha\beta)^3$$

$$= (-2\sqrt{2}(8 + 3))^2 + 2$$

$$= (8)(121) + 2 = 970$$

$$\frac{1}{10}(\alpha^6 + \beta^6) = 97$$

$$x^2 - (98 + 97)x + (98)(97) = 0$$

$$\Rightarrow x^2 - 195x + 9506 = 0$$

Q9

$$x^2 - \sqrt{2}x - \sqrt{3} = 0 \left\langle \begin{matrix} \alpha \\ \beta \end{matrix} \right.$$

$$\alpha^{n+2} - \sqrt{2}\alpha^{n+1} - \sqrt{3}\alpha^n = 0$$

$$\text{and } \beta^{n+2} - \sqrt{2}\beta^{n+1} - \sqrt{3}\beta^n = 0$$

Subtracting

$$(\alpha^{n+2} - \beta^{n+2}) - \sqrt{2}(\alpha^{n+1} - \beta^{n+1}) - \sqrt{3}(\alpha^n - \beta^n) = 0$$

$$\Rightarrow P_{n+2} - \sqrt{2}P_{n+1} - \sqrt{3}P_n = 0$$

Put $n = 10$

$$P_{12} - \sqrt{2}P_{11} - \sqrt{3}P_{10} = 0$$

$n = 9$

$$P_{11} - \sqrt{2}P_{10} - \sqrt{3}P_9 = 0$$

$$11(\sqrt{3} \cdot P_{10} + \sqrt{2}P_{11} - P_{11}) - 10(\sqrt{2}P_{10} - P_{11})$$

$$= 0 - 10(-\sqrt{3}P_9) = 10\sqrt{3}P_9$$

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